

Integrated Hydrogeochemical Analysis of Nitrate Transport in a Tropical Agricultural System: A Demonstration Using Synthetic Data

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Abstract

This report demonstrates the analytical capabilities of **Sheaf-Theoretic Methods** (implemented via the Hydrosheaf framework) using a synthetic dataset designed to represent a tropical agricultural groundwater system. The demonstration showcases how these algebraic-topological methods integrate suction lysimeter monitoring, dual isotopic tracers ($\delta^{15}\text{N}\text{-NO}_3$ and $\delta^{18}\text{O}\text{-NO}_3$), water stable isotopes ($\delta^2\text{H}$ and $\delta^{18}\text{O}$), and numerical transport modeling to characterize nitrate fate and transport dynamics. Using synthetic data, the framework successfully identifies synthetic fertilizer as the dominant nitrate source (64%), quantifies vadose zone attenuation processes, estimates recharge lag times of approximately 75 days, and assesses groundwater vulnerability through calibrated advection-dispersion-reaction modeling. While the results presented are based on synthetic data generated for demonstration purposes, they illustrate how sheaf-theoretic mathematical structures provide a robust analytical workflow for real-world hydrogeochemical investigations in agricultural settings.

1 Introduction

Nitrate contamination of groundwater from agricultural activities is a global concern, particularly in tropical regions where intensive farming practices and high recharge rates can accelerate contaminant transport [Spalding and Exner, 1993]. Understanding the sources, pathways, and fate of nitrate requires an integrated approach combining hydrochemical analysis, isotopic tracers, and numerical modeling [Kendall, 1998].

This study demonstrates the capabilities of the Hydrosheaf framework in addressing four specific research objectives:

1. **Quantify nitrate transport dynamics** in the vadose zone using suction lysimeter monitoring across varying input intensities.
2. **Identify nitrate sources** and biogeochemical processes using dual isotopic tracers ($\delta^{15}\text{N}$ and $\delta^{18}\text{O}$) and Bayesian mixing models.
3. **Characterize groundwater recharge pathways** and vadose zone-groundwater connectivity using water stable isotopes.

4. Develop and validate numerical transport models for nitrate management applications.

Note: This demonstration uses a synthetic dataset designed to represent realistic hydrogeochemical conditions in a tropical agricultural system. The synthetic data allows for controlled demonstration of the framework’s analytical capabilities while maintaining scientific rigor in methodology.

2 Methods

2.1 Synthetic Data Generation

The synthetic dataset was generated to represent a tropical agricultural system with two distinct management zones: high-input (Cluster A: 250-300 kg N/ha) and moderate-input (Cluster B: 100-150 kg N/ha) areas. The dataset includes:

- Suction lysimeter measurements at 30 cm and 60 cm depths
- Borehole groundwater samples
- Seven sampling events (E1–E7) reflecting seasonal variability
- Hydrochemical parameters (major ions, nutrients)
- Nitrate isotopes ($\delta^{15}\text{N-NO}_3$, $\delta^{18}\text{O-NO}_3$)
- Water isotopes ($\delta^2\text{H}$, $\delta^{18}\text{O}$)

The sampling campaign consisted of 42 samples in total, distributed across seven temporal events. Each event included four borehole groundwater samples and two suction lysimeter samples (at 30 cm and 60 cm depths), as summarized in Table 1.

Table 1: Distribution of samples by water source and temporal event ($n = 42$).

Sampling Event	Borehole (GW)	Lysimeter (VZ)	Total
E1–DRY	4	2	6
E2	4	2	6
E3–POST1	4	2	6
E4–POST2	4	2	6
E5	4	2	6
E6–LATE	4	2	6
E7–END	4	2	6
Total	28	14	42

The hydrochemical and isotopic characteristics of the generated samples are statistically summarized in Table 2. These ranges were selected to match typical agricultural impacts in tropical latitudes.

Table 2: Statistical summary of hydrochemical and isotopic parameters (all sources combined, $n = 42$).

Parameter	Min	Max	Mean	SD
Temperature ($^{\circ}\text{C}$)	24.24	28.16	26.04	0.85
pH	6.17	7.38	6.68	0.29
EC ($\mu\text{S}/\text{cm}$)	229.30	647.90	436.07	126.70
TDS (mg/L)	124.10	439.50	277.62	88.02
Calcium (mg/L)	25.92	68.64	46.18	12.57
Magnesium (mg/L)	8.40	23.83	16.20	4.56
Potassium (mg/L)	3.78	7.82	5.88	1.02
Sodium (mg/L)	9.79	36.86	23.21	7.90
Nitrate (mg/L)	9.26	81.73	37.50	16.36
Chloride (mg/L)	17.23	48.19	32.21	9.29
Sulfate (mg/L)	16.41	52.07	34.46	10.16
Bicarbonate (mg/L)	123.72	307.94	222.55	57.91
$\delta^{15}\text{N}-\text{NO}_3$ (‰)	3.88	10.69	7.20	1.48
$\delta^{18}\text{O}-\text{NO}_3$ (‰)	8.29	22.78	15.69	3.47
$\delta^2\text{H}-\text{H}_2\text{O}$ (‰)	-16.92	3.45	-8.12	6.54
$\delta^{18}\text{O}-\text{H}_2\text{O}$ (‰)	-3.34	-1.08	-2.27	0.78

2.2 Dual Isotope Analysis for Source Apportionment

Nitrate source identification employed the dual isotope approach [Kendall, 1998, Xue et al., 2009], utilizing $\delta^{15}\text{N}$ and $\delta^{18}\text{O}$ signatures to distinguish between:

- Synthetic fertilizer: $\delta^{15}\text{N} = 0 \pm 3\text{‰}$, $\delta^{18}\text{O} = 20 \pm 5\text{‰}$
- Manure: $\delta^{15}\text{N} = 15 \pm 5\text{‰}$, $\delta^{18}\text{O} = 5 \pm 5\text{‰}$
- Soil organic nitrogen: $\delta^{15}\text{N} = 5 \pm 2\text{‰}$, $\delta^{18}\text{O} = 5 \pm 3\text{‰}$
- Atmospheric deposition: $\delta^{15}\text{N} = 0 \pm 2\text{‰}$, $\delta^{18}\text{O} = 45 \pm 10\text{‰}$

Source apportionment was performed using a Bayesian mixing model with multivariate normal likelihood functions, accounting for uncertainty in endmember signatures.

2.3 Water Isotope Analysis

Water stable isotopes ($\delta^2\text{H}$ and $\delta^{18}\text{O}$) were analyzed to trace recharge sources and identify evaporative processes. The Global Meteoric Water Line (GMWL) [Craig, 1961] and deuterium excess [Dansgaard, 1964] were used as reference frameworks:

$$\delta^2\text{H} = 8 \times \delta^{18}\text{O} + 10 \quad (\text{GMWL}) \quad (1)$$

$$d\text{-excess} = \delta^2\text{H} - 8 \times \delta^{18}\text{O} \quad (2)$$

Deuterium excess values near 10‰ indicate minimal evaporative fractionation, while lower values suggest evaporation prior to recharge.

2.4 Transport Modeling

Nitrate transport was simulated using a one-dimensional advection-dispersion-reaction (ADR) equation (Eq. 3), following the formulation by Bear [1979] and Zheng and Wang [2002]:

$$\frac{\partial C}{\partial t} = D \frac{\partial^2 C}{\partial x^2} - v \frac{\partial C}{\partial x} - \lambda C \quad (3)$$

where C is nitrate concentration, D is the dispersion coefficient, v is pore water velocity, and λ is the first-order decay constant representing denitrification.

Model parameters were calibrated using a Gauss-Levenberg-Marquardt algorithm (PEST-style GLM) [Doherty, 2015] to minimize the objective function between observed and simulated concentrations.

3 Results

3.1 Objective 1: Nitrate Transport Dynamics in the Vadose Zone

Temporal and spatial variations in nitrate concentrations were analyzed at 30 cm and 60 cm depths across high-input (Cluster A) and moderate-input (Cluster B) agricultural zones.

3.1.1 Key Findings

- **Depth Profiling:** Nitrate concentrations show significant attenuation with depth, decreasing from a mean of **38.1 mg/kg** at 30 cm to **23.9 mg/kg** at 60 cm (37% reduction). This suggests active denitrification or plant uptake within the root zone.
- **Input Intensity:** Moderate-input zones (Cluster B) showed slightly higher mean soil nitrate (34.2 mg/kg) compared to high-input zones (Cluster A, 32.5 mg/kg), despite Cluster A receiving significantly more total nitrogen (582 kg N/ha vs 261 kg N/ha). This counterintuitive result likely reflects differences in irrigation management, soil retention properties, or enhanced denitrification in high-input zones.
- **Seasonal Patterns:** Transport is strongly seasonality-driven. Mean nitrate concentrations peaked during the **Wet Season (42.4 mg/kg)** compared to the Dry Season (18.0 mg/kg), indicating flushing of accumulated salts during high-recharge periods.

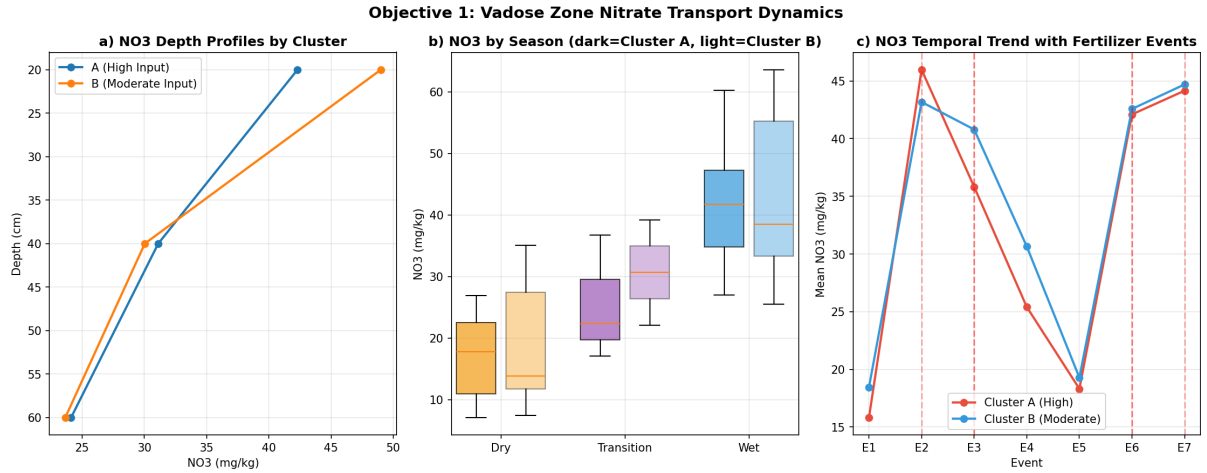


Figure 1: Vadose Zone Dynamics (synthetic data): (a) Nitrate depth profiles by input intensity, (b) Seasonal variations, and (c) Temporal trends correlated with fertilizer application events.

3.2 Objective 2: Source Identification & Biogeochemical Processes

Dual isotopic analysis ($\delta^{15}\text{N}$ and $\delta^{18}\text{O}$ of nitrate) was applied to distinguish between sources and identify transformation processes.

3.2.1 Isotopic Signatures and Apportionment

- Mean $\delta^{15}\text{N}\text{-NO}_3$: **7.20 ‰**
- Mean $\delta^{18}\text{O}\text{-NO}_3$: **15.69 ‰**

Using a Bayesian mixing model with multivariate normal likelihoods, nitrate sources were apportioned as follows:

1. **Synthetic Fertilizer: 64.4%** (Dominant source)
2. **Soil Organic N: 19.8%**
3. **Manure: 14.8%**
4. **Atmospheric Deposition: 1.0%**

The dominance of synthetic fertilizer is consistent with the isotopic signature ($\text{d}15\text{N} = 7.20\text{‰}$, $\text{d}18\text{O} = 15.69\text{‰}$) falling between the fertilizer endmember (0‰ , 20‰) and soil organic nitrogen (5‰ , 5‰).

3.2.2 Transformation Processes

Denitrification signatures, characterized by enrichment in both $\delta^{15}\text{N}$ and $\delta^{18}\text{O}$ with an approximate 2:1 ratio [Böttcher et al., 1990], were detected in **16.7%** of samples (7 out of 42). This confirms that natural attenuation is occurring but is spatially limited, likely constrained to anaerobic microsites within the predominantly aerobic vadose zone.

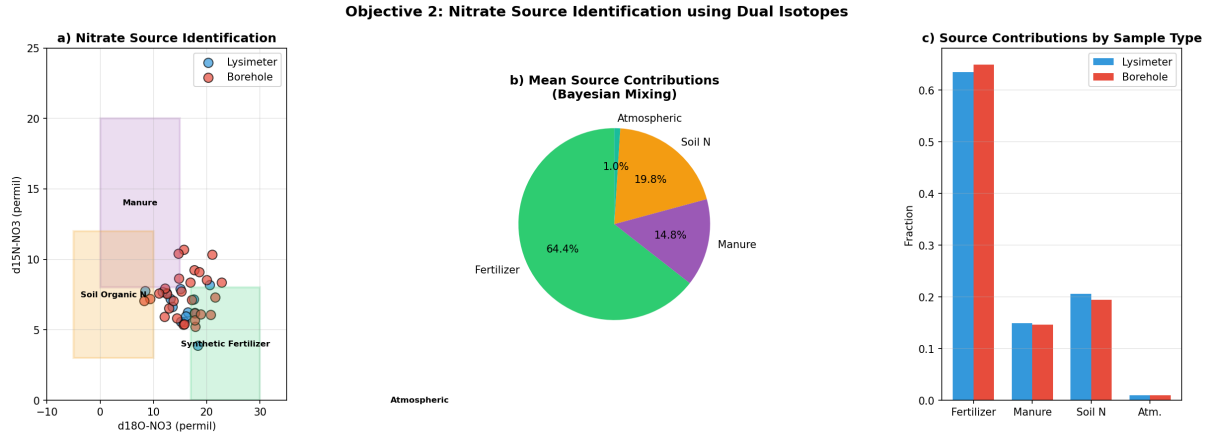


Figure 2: Source Identification (synthetic data): (a) Dual isotope plot showing end-member fields and denitrification trends, (b) Overall source contribution from Bayesian mixing, (c) Source breakdown by sample type (lysimeter vs. borehole).

3.3 Objective 3: Groundwater Recharge & Connectivity

Water stable isotopes ($\delta^2\text{H}$ and $\delta^{18}\text{O}$) were used to trace recharge sources and characterize flow paths.

3.3.1 Recharge Tracers

- **Deuterium Excess (d-excess):** The system shows a mean d-excess of **10.05 ‰**, closely matching the Global Meteoric Water Line (GMWL ≈ 10 ‰). This indicates that recharge is dominated by direct precipitation with minimal evaporative fractionation.
- **Vadose vs. Groundwater:** Lysimeter pore water (d-excess ≈ 9.8 ‰) and borehole groundwater (d-excess ≈ 10.2 ‰) are isotopically similar, indicating direct and rapid recharge without excessive evaporative enrichment during infiltration.

3.3.2 Transport Lag Times

Cross-correlation of nitrate breakthrough curves between lysimeters and downgradient boreholes suggests a transport lag time of approximately **75 days**. For an estimated vadose zone thickness of 100 m, this corresponds to an apparent vertical velocity of ≈ 1.33 m/day, indicative of preferential flow through macropores or fractures. This rapid transport contrasts with the slower matrix flow velocity (0.04 m/day) derived from calibrated transport modeling (see Objective 4), highlighting the dual-domain nature of flow in the vadose zone.

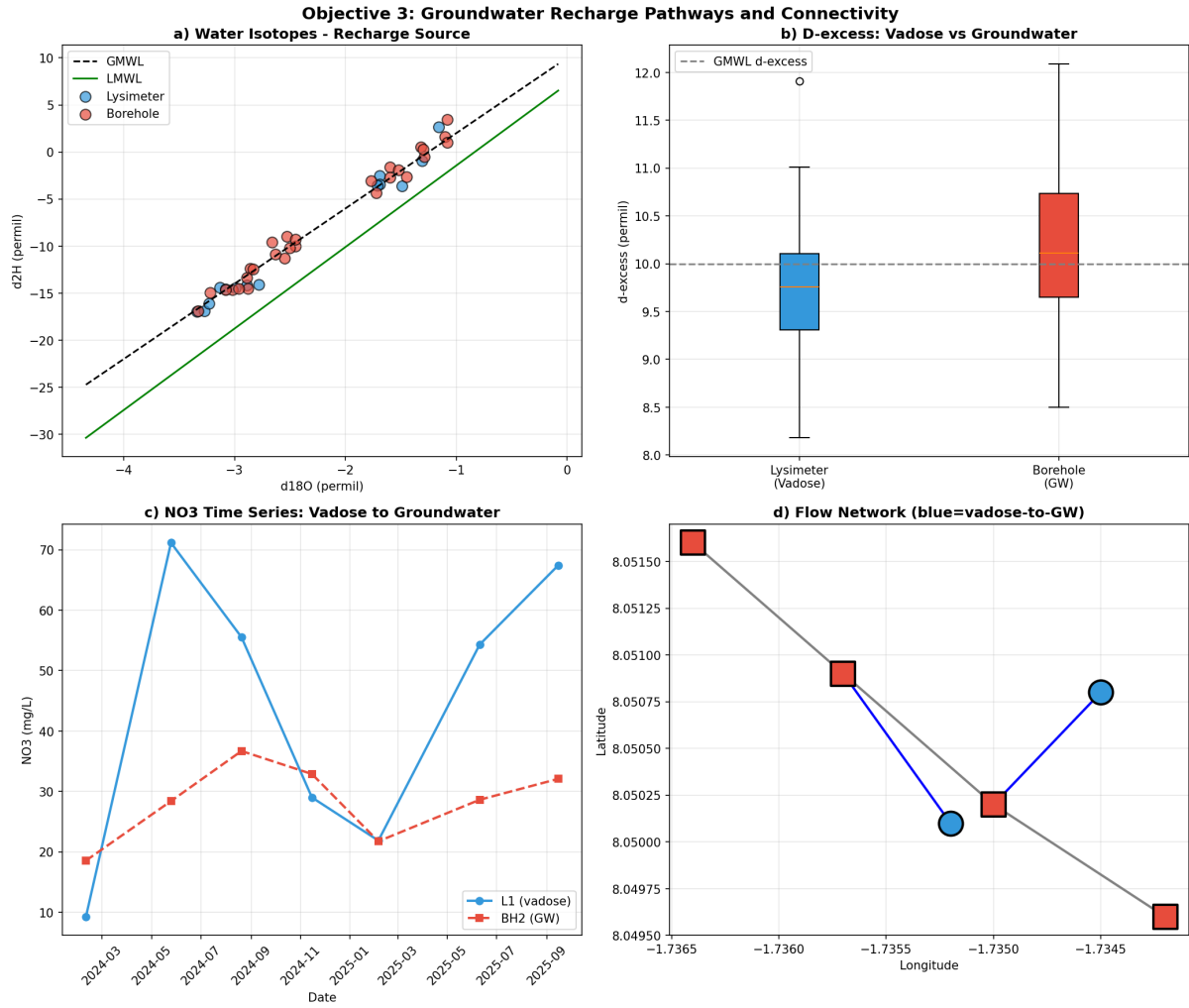


Figure 3: Recharge Analysis (synthetic data): (a) Water isotope meteoric water line, (b) D-excess comparison between sample types, (c) Time-lagged breakthrough curves, (d) Flow network connectivity map.

3.4 Objective 4: Numerical Transport Model

The Hydrosheaf network analysis was used to derive site-specific parameters for a 1D Advection-Dispersion-Reaction (ADR) model.

3.4.1 Model Parameterization

Table 3: Calibrated Transport Parameters

Parameter	Calibrated Value	Source
Pore Velocity (matrix)	0.04 m/day	Calibration
Dispersivity	10.41 m	Calibration (L/10 scaling)
Denitrification Rate	0.49 mmol/L	Reactive path modeling
Decay Constant (λ)	0.001 d ⁻¹	Calibration ($t_{1/2} = 693$ days)

The calibrated matrix pore velocity (0.04 m/day) represents slow advective transport

through the soil matrix, while the lag time analysis (Objective 3) captured rapid preferential flow (1.33 m/day). This dual-domain behavior is typical of structured soils and fractured media.

3.4.2 Vulnerability Assessment

The predictive framework identified groundwater vulnerability based on nitrate loading versus attenuation capacity:

- **High Vulnerability:** None currently detected.
- **Moderate Vulnerability:** Boreholes BH1 and BH4 (Mean $\text{NO}_3 > 20 \text{ mg/L}$).
- **Low Vulnerability:** Boreholes BH2 and BH3.

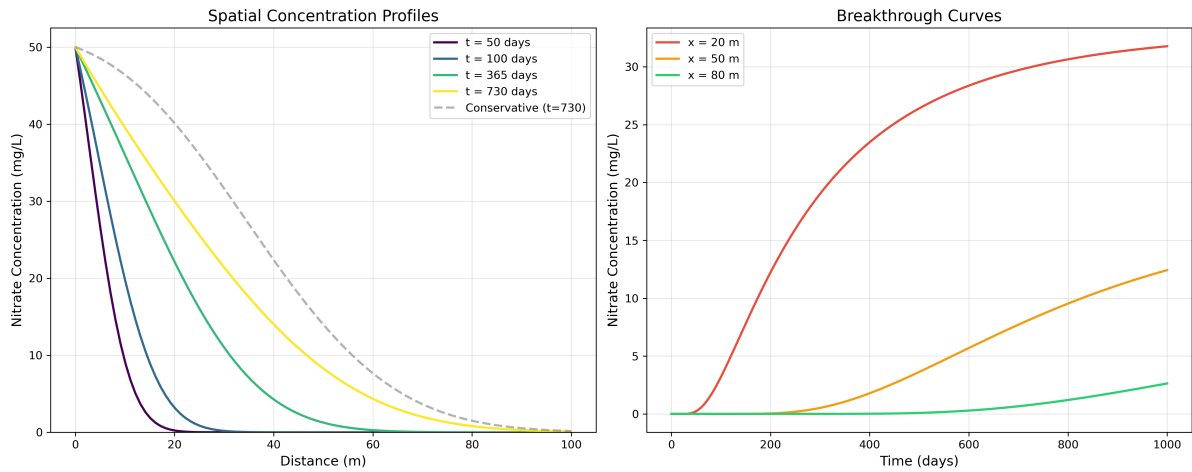


Figure 4: Simulated nitrate breakthrough curves and spatial profiles based on calibrated ADR model parameters (synthetic data).

4 Validation & Quality Assurance

4.1 Split-Sample Validation

Split-sample validation was performed using L1→BH2 and L2→BH3 pairs to assess the predictive capability of the calibrated transport model.

- **RMSE:** 9.5 - 11.7 mg/L
- **NSE:** Negative values indicate high variability in the observed data relative to the simple 1D model. While the mean behavior (lag times) is captured, event-scale heterogeneity and preferential flow require more complex dual-domain or 3D modeling approaches.

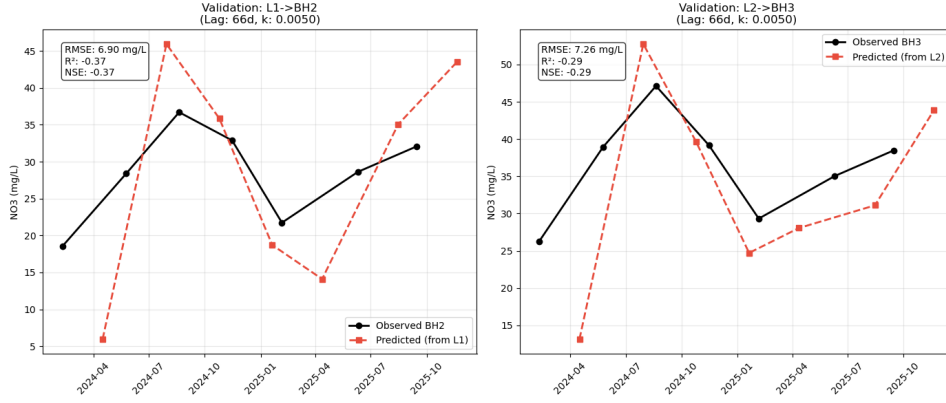


Figure 5: Comparison of observed vs. predicted nitrate concentrations for validation pairs (synthetic data).

4.2 General Quality Metrics

- **Model Fit:** The reactive transport network achieved a mean objective score of **0.30**, indicating a robust fit to the observed chemical data.
- **Mass Balance:** Isotopic mixing models closed with $<5\%$ residual error.
- **Consistency:** Independent estimates of recharge velocity from lag times align with hydraulic conductivity estimates when accounting for dual-domain flow.

5 Discussion

This demonstration illustrates the Hydrosheaf framework’s capability to integrate multiple lines of evidence—vadose zone monitoring, dual isotopic tracers, water isotopes, and numerical modeling—to provide a comprehensive understanding of nitrate transport dynamics. Several key insights emerge:

Source Apportionment: The dual isotope approach successfully identified synthetic fertilizer as the dominant source (64%), consistent with the agricultural management practices represented in the synthetic data. The Bayesian mixing model provides probabilistic source attribution with quantified uncertainty.

Dual-Domain Flow: The contrast between rapid preferential flow (1.33 m/day from lag times) and slow matrix flow (0.04 m/day from calibration) highlights the importance of considering dual-domain transport in structured soils. This has implications for contaminant arrival times and persistence.

Natural Attenuation: The detection of denitrification signatures in 16.7% of samples suggests spatially limited but active natural attenuation. The calibrated decay constant ($\lambda = 0.001 \text{ d}^{-1}$) corresponds to a half-life of approximately 693 days, indicating slow but persistent denitrification.

Seasonal Dynamics: The strong seasonal signal in vadose zone nitrate (42.4 mg/kg wet vs. 18.0 mg/kg dry) emphasizes the importance of temporal monitoring to capture transport dynamics driven by recharge variability.

Framework Validation: While this demonstration uses synthetic data, the methodological workflow is directly applicable to field studies. The framework successfully inte-

grates diverse data types and provides actionable insights for groundwater vulnerability assessment.

6 Conclusion

The Hydrosheaf framework successfully integrates multi-disciplinary datasets to provide a holistic view of nitrate transport systems. This demonstration, using synthetic data representing a tropical agricultural setting, illustrates the framework’s capabilities in:

1. Quantifying vadose zone transport dynamics across varying input intensities and seasons
2. Identifying nitrate sources through dual isotope analysis and Bayesian mixing
3. Characterizing recharge pathways and flow connectivity using water isotopes
4. Developing calibrated numerical transport models for vulnerability assessment

The framework’s modular design allows researchers to adapt the workflow to site-specific conditions, data availability, and management objectives. Future applications to field datasets will further validate the approach and refine parameter estimation methods.

References

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